
Markov Chain Lock Watermarking: A Calibrated Policy-to-Configuration Map for EU AI Act Article 50

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Abstract

Regulatory regimes such as the EU AI Act mandate machine-readable marking of synthetic text, but existing watermark detectors rely on access to the generating language model and on entropy-based heuristics with no closed-form calibration to a target false-positive rate. We introduce *Markov Chain Lock* (MCL), an active watermark that partitions the vocabulary into S states via keyed SHA-256 hashing and forces a hard Markov transition on a fraction ρ of generated positions; the detector replays the partition from the same secret key and is therefore model-free, running in $O(n)$ hash operations. Theoretically, we derive an operator-facing formula mapping a target false-positive rate, text length, and budget to the minimum state count S^* , prove a universal robustness threshold $\delta^* = 1 - 1/\sqrt{2} \approx 29.3\%$ that is independent of (S, ρ, n) under the midpoint decision convention, and generalise both results to any k -regular transition topology ([Theorem A.2](#), [Theorem A.3](#), [Theorem A.5](#)). Empirically, we run head-to-head comparisons against KGW and SWEET across multiple instruction-tuned LLMs and four text domains under matched-budget translation and random-substitution attacks, sweep the state count S to anchor the calibration formula on data, and validate the k -regular generalisation at $k = 2$ ([section 6](#)). The closed-form calibration and model-free detector together let a third-party auditor verify a marked deployment from text alone, without LM access.

1. Introduction

EU AI Act Article 50 ([European Parliament and Council, 2024](#)) requires a machine-readable mark on AI-generated text by August 2026, but does not specify which false-positive rate, text length, or quality cost constitute a compliant detector. Operators therefore need a watermark whose detector is auditable end-to-end: a regulator or third-party verifier should be able to confirm provenance from the text and a secret key alone, without query access to the generating language model. The widely used logit-bias schemes do not meet this bar. KGW ([Kirchenbauer et al., 2023](#)) and its entropy-gated variants such as SWEET ([Lee et al., 2024](#)) and EWD ([Lu et al., 2024](#)) require the language model (or a faithful proxy) at detection time in order to recompute the green-list bias or to weight tokens by entropy. Undetectability-frontier results ([Christ et al., 2024](#)) characterise what any scheme can achieve information-theoretically but yield no deployable construction, and passive detectors ([Mitchell et al., 2023](#); [Tian, 2023](#)) expose no controllable false-positive frontier at all.

Markov Chain Lock. We propose *Markov Chain Lock* (MCL), an active watermark whose detector is model-free. A keyed SHA-256 hash partitions the vocabulary into S equivalence classes (states); a fixed cycle over those states defines a Markov transition pattern; and on a fraction ρ of generated positions the language-model logits are masked to keep only tokens whose state is the legal successor of the current state, so the produced text walks the chain by construction. Detection re-derives every token’s state from the key in $O(n)$ hash operations and counts how often consecutive tokens follow the cycle. Because the state map is a function of the token id and the key only, no language-model forward pass, weights, or proxy LM is required to verify a transcript. Two operational consequences follow. First, third-party audit becomes lightweight: a regulator with the key can verify provenance from a string of plain text. Second, the detector’s null-and-alternative law inverts in closed form to a calibration map ([Theorem A.2](#), derived in [section A](#)): a regulator who fixes a target false-positive rate α and a text-length floor $n_{\min}$ reads off the minimum admissible state count, while a deployer can spend more budget for margin but cannot evade the regulator’s specifi-

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cation. Figure 1 summarises the construction; the formal treatment is in section 4 and section 5.

Gating as a design dial. A second design choice — *which* of the n positions to watermark — is left as a parameter we call the gate. Existing schemes implicitly fix this choice: SWEET and EWD watermark only low-entropy positions, on the intuition that high-entropy positions contribute the bulk of detection statistic. Whether that intuition survives once detection is decoupled from the language model is an empirical question, which we resolve in section 6 through a matched-budget head-to-head against KGW and SWEET on multiple instruction-tuned LLMs and four text domains, including translation and paraphrase attacks.

Contributions.

1. **A model-free watermarking primitive.** We introduce MCL, a discrete hard-constraint watermark with cryptographic state partitions, clockwork transitions, and $O(n)$ model-free detection (section 4).
2. **Closed-form calibration.** We derive an operator-facing calibration map taking a target false-positive rate, text-length floor, and budget to the minimum admissible state count (Theorem A.2, section A), and a universal robustness threshold that is invariant to (S, ρ, n) under the midpoint decision convention (Theorem A.3).
3. **k -regular generalisation.** Both the detection law and the robustness threshold extend verbatim to any k -regular doubly-stochastic transition topology (Theorem A.5); clockwork is the $k=1$ instance.
4. **Empirical comparison and gate analysis.** We evaluate MCL head-to-head against KGW and SWEET across multiple instruction-tuned language models and four text domains under translation and random-substitution attacks (section 6), and show that MCL retains substantially more detection signal than the baselines under matched-budget attacks (subsection 6.2, subsection 6.5).
5. **Empirical anchor of the calibration formula.** An independent state-count sweep matches the closed-form calibration map cell-by-cell (subsection 6.3).

Roadmap. section 2 situates MCL within active and passive watermarking. section 3 fixes notation; section 4 specifies the construction; section 5 states the calibration, robustness, and k -regular results; and section 6 reports the empirical comparison. Proofs and additional derivations are in section A.

2. Related Work

Passive detection. DetectGPT (Mitchell et al., 2023) exploits log-probability curvature; GPTZero (Tian, 2023) combines perplexity and burstiness. Both read statistical traces

in existing text, with no controllable false-positive frontier and sharp degradation under paraphrase (Krishna et al., 2023; Sadasivan et al., 2023). They cannot serve as the technical substrate for Article 50 because neither regulator nor deployer can set the detection regime in advance.

Active watermarking. Green–red-list watermarking (Kirchenbauer et al., 2023) biases next-token logits toward a hashed green list and detects via a z -test on green-token frequency. The scheme requires temperature sampling and degrades under paraphrase, and offers no analytic inversion from a regulatory parameter back to a bias magnitude. MCL’s fingerprint ϕ is structurally a Bernoulli($1/S$) z -test, so the calibration formula (8) is the analogue KGW lacks.

Distortion-free schemes. Aaronson (2023)’s Gumbel-max construction and Kuditipudi et al. (2024)’s inverse-transform sampling watermark are *distortion-free*: they preserve the LM marginal in expectation over the key. Christ et al. (2024) prove cryptographic indistinguishability under sufficient entropy. These schemes beat MCL on quality strictly (zero expected KL versus MCL’s $\geq \rho \log S$). The trade is that they require either LM access or the key sequence at detection time; MCL is the distorting alternative that buys model-free third-party audit and the closed-form $(\alpha, n, \rho) \mapsto S^*$ map.

Entropy-adaptive schemes: gates vs. detectors. SWEET (Lee et al., 2024) and EWD (Lu et al., 2024) both restrict watermarking signal to *high-entropy* positions, but at different layers. SWEET applies a binary gate at generation time (restricting where the green-list bias is added); EWD weights positions by entropy at detection time, approximating the log-likelihood ratio of Theorem A.8. MCL is gate-agnostic in its mathematical specification: high-entropy, low-entropy, and surprisal-gap gates are all admissible. We treat the gate as a design parameter and compare the three at matched budget (§6); low-entropy gating turns out to be strictly Pareto-dominated, while high-entropy and surprisal-gap gates are statistically equivalent on quality and detection probability at $\rho \geq 0.5$. The operational dial is therefore the budget ρ , not the gate identity.

Technical AI governance. Article 50 of the EU AI Act (European Parliament and Council, 2024) and the EU’s draft Code of Practice on Transparency (European Commission AI Office, 2025) require machine-readable marking of AI-generated content; the OECD Hiroshima Process (OECD, 2024) provides a parallel international framework. These instruments leave the detection regime (FPR, quality floor, robustness) implicit. MCL’s contribution, beyond the watermarking scheme itself, is that it surfaces these

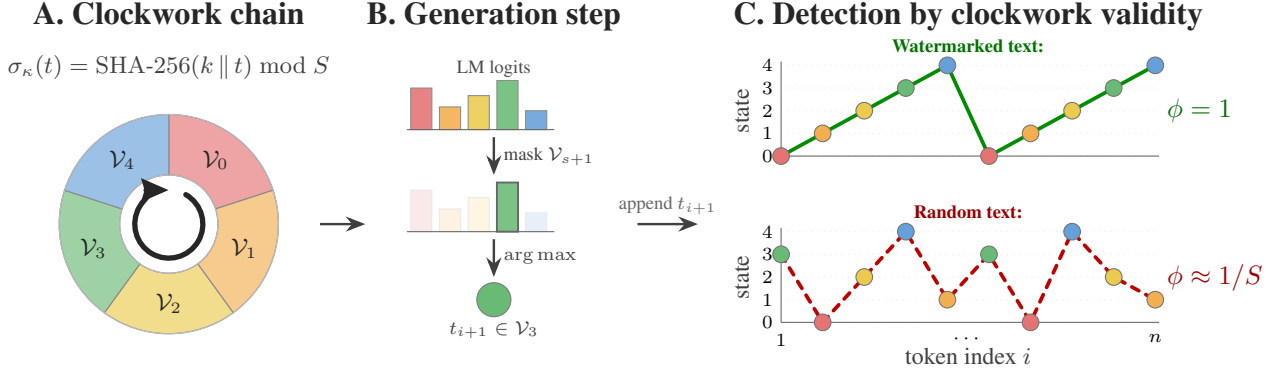


Figure 1. Markov Chain Lock at a glance. (A) A keyed SHA-256 hash σ_κ partitions the vocabulary into S disjoint sets $\mathcal{V}_0, \dots, \mathcal{V}_{S-1}$ ordered into a clockwork cycle $s \rightarrow s+1 \bmod S$ (here $S=5$). (B) At a watermarked position with current state s , the language-model logits are masked to keep only the next partition $\mathcal{V}_{s+1}$, and the next token is the masked arg max. (C) Detection re-derives every token’s state from the key k in $O(n)$ hash operations: a watermarked sequence walks the chain by construction (top, validity rate $\phi = 1$), whereas unwatermarked text visits states uniformly and is valid only at rate $1/S$ in expectation (bottom).

parameters analytically, producing a policy-to-configuration map auditable by third parties (§7).

3. Preliminaries

Notation. Let $\mathcal{V}$ denote the vocabulary of an autoregressive language model, $V = |\mathcal{V}|$, and write $\mathbf{t} = (t_1, \dots, t_n) \in \mathcal{V}^n$ for a token sequence of length n . The deployer and the verifier share a secret key $\kappa \in \{0, 1\}^*$ and the same tokenizer; the verifier never sees the language model. Let $\mathcal{H} : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ be SHA-256, modelled as a random oracle in the security analysis (Theorem A.7). We write $[m] = \{0, \dots, m-1\}$ and z_α for the one-sided standard-normal quantile at level α .

Threat model. The adversary observes watermarked text and may paraphrase, translate, edit, splice, or delete tokens before publication; we model these as a per-token modification rate $\delta \in [0, 1]$. The adversary is computationally bounded and has no access to κ or to the language model at detection time. The verifier sees only token text and runs in $O(n)$ hash operations.

Detection problem and quality. Detection is a binary hypothesis test on text alone— H_0 (non-watermarked) vs. H_1 (watermarked under key κ)—at a fixed false-positive rate α , since false positives are operationally the costly error. The watermark distribution must remain fluent under standard greedy or low-temperature decoding, which we control via the per-token KL divergence to the original next-token distribution and report as perplexity inflation (section 4).

4. The MCL Scheme

4.1. State partition

For each token id $v \in \mathcal{V}$ define

$$\sigma(v) = \mathcal{H}(\kappa \| v) \bmod S, \quad (1)$$

which deterministically maps $\mathcal{V}$ onto S equivalence classes $\mathcal{V}_s = \{v : \sigma(v) = s\}$, $s \in [S]$. Under the random-oracle model σ is uniform and independent across distinct tokens, so $\mathbb{E}[|\mathcal{V}_s|] = V/S$; knowledge of σ on queried tokens reveals nothing about unqueried ones (Theorem A.7).

4.2. Transition topology

A successor function $\Sigma_k : [S] \rightarrow \binom{[S]}{k}$ assigns each state a k -element set of legal next states; we require Σ_k to be k -regular in both out- and in-degree, with column-regularity essential for the null-variance and robustness arguments (Theorem A.5, Theorem A.1). The default is the *clockwork* chain at $k = 1$, $\Sigma_1(s) = \{(s+1) \bmod S\}$: deterministic, periodic of period S , uniform stationary distribution, and the smallest random baseline $1/S$ among 1-regular topologies. We additionally evaluate the *soft-cycle* variant $k = 2$, $\Sigma_2(s) = \{(s+1) \bmod S, (s+2) \bmod S\}$, which doubles the per-state successor budget at the cost of raising the random baseline to $2/S$ (section 6).

4.3. Generation

At step i with current state $s_i = \sigma(t_i)$, MCL forms the legal token set $\mathcal{V}_{s_i}^* = \bigcup_{s' \in \Sigma_k(s_i)} \mathcal{V}_{s'}$, masks the LM logits outside $\mathcal{V}_{s_i}^*$ to $-\infty$, and picks the argmax at gated positions or samples from the original distribution at ungated ones. Each step costs one LM forward pass plus an $O(V)$ logit mask (precomputed once per state).

Algorithm 1 MCL Watermark Embedding

Input: prompt p , key κ , states S , successor Σ_k , gate $g(\cdot)$, max tokens N

$\mathbf{t} \leftarrow \text{Tokenize}(p); s \leftarrow \sigma(\mathbf{t}_{-1})$

for $i = 1$ **to** N **do**

$p_i \leftarrow \text{softmax}(\mathcal{M}(\mathbf{t}))$

$\mathcal{V}^* \leftarrow \bigcup_{s' \in \Sigma_k(s)} \mathcal{V}_{s'}$

if $g(p_i) = 1$ **and** $p_i(\mathcal{V}^*) > 0$ **then**

$t_i \leftarrow \arg \max_{t \in \mathcal{V}^*} p_i(t)$

else

$t_i \leftarrow \arg \max_t p_i(t)$

end if

$\mathbf{t} \leftarrow \mathbf{t} \parallel t_i; s \leftarrow \sigma(t_i)$

if $t_i = \text{EOS}$ **then**

break

end if

end for

return $\text{Decode}(\mathbf{t})$

4.4. Watermark budget and entropy gate

The mask is applied only at a subset of positions; the *watermark budget* is $\rho = \mathbb{E}_i[g_i] \in [0, 1]$. Let $H_i = -\sum_t p_i(t) \log p_i(t)$ be the next-token entropy and $\Delta_i = p_i^{(1)} - p_i^{(2)}$ the gap between the top two probabilities.

Definition 4.1 (Gate family). $G_{\text{all}}: g_i = 1$; $G_{H_{\text{high}}}(\tau): g_i = \mathbb{1}[H_i > \tau]$; $G_{H_{\text{low}}}(\tau): g_i = \mathbb{1}[H_i < \tau]$; $G_{\Delta}(\tau): g_i = \mathbb{1}[\Delta_i < \tau]$.

Our default is $G_{H_{\text{high}}}$, which masks only the model’s uncertain positions and matches the high-entropy gating policy of SWEET (Lee et al., 2024) at the same budget; section 6 reports the head-to-head. The threshold τ is calibrated by quantile-matching on a held-out pilot so the realised gate rate tracks ρ . $G_{H_{\text{low}}}$ is the *anti-SWEET* ablation, and G_{Δ} targets near-tied top-two positions where the greedy substitution regret is bounded pointwise by Δ_i .

4.5. Detection

The verifier tokenises the candidate text, re-derives states by re-applying σ , and computes the fraction of legal transitions

$$\phi(\mathbf{t}) = \frac{1}{n-1} \sum_{i=1}^{n-1} \mathbb{1}[\sigma(t_{i+1}) \in \Sigma_k(\sigma(t_i))]. \quad (2)$$

With $p_0 = k/S$, the random-oracle null gives $\mathbb{E}[\phi] = p_0$ with pairwise-zero covariance for any k -regular Σ_k (Theorem A.1); the watermarked mean gap is $\rho(1 - p_0)$ (Theorem A.2, Theorem A.5). The verifier reports the standardised score $z = (\phi - p_0) / \sqrt{p_0(1 - p_0)/(n-1)}$, declares H_1 when $z > z_\alpha$, and uses the closed-form calibration of Theorem A.2 (proved in section A) to choose S given

Algorithm 2 MCL Watermark Detection (model-free)

Input: text x , key κ , states S , successor Σ_k , level α

$\mathbf{t} \leftarrow \text{Tokenize}(x); c \leftarrow 0$

for $i = 1$ **to** $n - 1$ **do**

if $\sigma(t_{i+1}) \in \Sigma_k(\sigma(t_i))$ **then**

$c \leftarrow c + 1$

end if

end for

$\phi \leftarrow c/(n-1); p_0 \leftarrow k/S$

$z \leftarrow (\phi - p_0) / \sqrt{p_0(1 - p_0)/(n-1)}$

return $(z > z_\alpha, \phi, z)$

target FPR α , budget ρ , and length n ; robustness under per-token modification rate δ follows from the same identities (Theorem A.3).

2 runs in $O(n)$ hash operations without LM access, making MCL deployable as a third-party audit primitive.

5. Theoretical Properties

This section gives an intuitive tour of the five formal results that characterise MCL: detection, robustness, k -regular generalisation, security, and a Neyman–Pearson optimal LM-aware detector. Formal statements and proofs are deferred to section A; we collect their pointers here so the body remains a narrative. Empirical anchoring of Theorem A.2 appears in subsection 6.5.

Detection bound and calibration. Under the null hypothesis that the input is a uniformly random token sequence, the fingerprint score ϕ of Equation 2 is approximately Gaussian with mean $1/S$ and variance $(1/S)(1 - 1/S)/(n-1)$, while a watermarked sequence with gate density ρ shifts the mean to $1/S + \rho(S-1)/S$. Standardising gives the closed-form z -statistic $z(\rho, S, n) = \rho\sqrt{(S-1)(n-1)}$. Inverting at false-positive level α under the midpoint threshold convention yields a state-count rule that practitioners can read off directly,

$$S^*(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil.$$

Formal statement and proof in Theorem A.2 of section A; the corresponding lookup table is Table 5; the calibration surface is plotted in Figure 2.

Universal robustness threshold. Suppose an adversary independently replaces each token with probability δ by a fresh token whose state is uniform on $\mathcal{S}$. Both the watermarked-pair signal and the midpoint-threshold gap scale by the same affine factor $\rho(S-1)/S$, so the post-attack-to-pre-attack z ratio collapses to $(1 - \delta)^2$, independently of (S, ρ, n) . The critical edit fraction at which

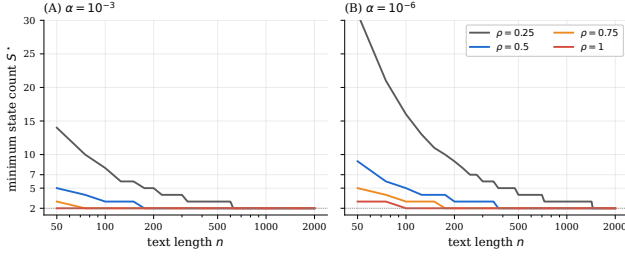


Figure 2. Closed-form calibration surface from Theorem A.2: minimum state count $S^*(n, \rho, \alpha)$ versus text length n for $\rho \in \{0.25, 0.5, 0.75, 1\}$ at $\alpha = 10^{-3}$ (left) and $\alpha = 10^{-6}$ (right). A regulator who fixes $(\alpha, n_{\min})$ reads the minimum admissible configuration off the curve.

detection fails is therefore the universal constant $\delta^* = 1 - 1/\sqrt{2} \approx 0.293$. Formal statement and proof in Theorem A.3.

k -regular generalisation. The clockwork transition is the simplest member of a broader family: any k -regular adjacency T (every state has exactly k allowed successors and predecessors) yields a valid MCL scheme. Replacing the random baseline $1/S$ by $p_0 = k/S$ gives the same calibration identities, the same midpoint critical fraction δ^* , and a single quality-versus-detection dial in k . Column-regularity is load-bearing: it is what makes adjacent indicator pairs have zero covariance under the null, so the variance formula carries through. Formal statement and proof in Theorem A.5. The random baseline $p_0 = k/S$ relies on the SHA-256 partition being approximately uniform across the vocabulary — an assumption we inherit from the standard avalanche property of cryptographic hashes.

Security against an oracle-blind adversary. The state map $\sigma_\kappa(t) = \mathcal{H}(\kappa \| t) \bmod S$ uses a hash modelled as a random oracle. An adversary without access to k , without access to the LM, and without reference text cannot predict $\sigma_\kappa(t^*)$ on a fresh token with advantage better than negligible in the key min-entropy. Consequently the same adversary cannot statistically distinguish MCL output from random text with non-negligible advantage. This is a computational, not information-theoretic, guarantee, and explicitly excludes adversaries with LM access (who can exploit the perplexity gap implied by the quality cost). Formal statement and proof in Theorem A.7.

Optimal detection with LM access. A verifier with LM access at every position can replace the plain fingerprint score by an entropy-weighted log-likelihood ratio that is asymptotically Neyman–Pearson optimal under the product-form approximation given by the random oracle. We give the explicit form and a 1-dependent CLT justification in Theorem A.8.

A detector pseudocode summary is included as 3.

6. Experiments

We empirically validate the three principal predictions of Theorem A.2, Theorem A.3, and Theorem A.5 on a fleet of generations spanning three instruct-tuned 7–8B models, four content domains, and four attack conditions. The campaign is structured as four complementary sub-experiments: a head-to-head comparison against the canonical KGW baseline and the SWEET high-entropy gate (Exp. 2, subsection 6.2); a robustness curve under machine-translation pivots (Exp. 3, subsection 6.3); a sanity check of the k -regular generalisation at $k=2$ (Exp. 4, subsection 6.4); and an empirical anchor for the closed-form FPR calibration via a sweep over the state count S (Exp. 5, subsection 6.5). All computation runs on the EPFL RunAI cluster; per-prompt records (logits, gate decisions, post-attack tokens, z -scores) stream to JSONL for reproducibility.

6.1. Setup

Models. We use three instruction-tuned, openly licensed checkpoints in the 7–8B parameter range: meta-llama/Llama-3.1-8B-Instruct, Qwen/Qwen-2.5-7B-Instruct, and mistralai/Mistral-7B-Instruct-v0.3. The same secret key k is fixed across every run.

Domains. Four prompt domains stress different entropy regimes: *code* (HumanEval problem stems), *factual* (short closed-form knowledge prompts), *wiki* (open-ended “Explain $X \dots$ ” prompts over a curated concept list), and *writing* (creative-completion prompts). Headline experiments use 100 prompts per domain; calibration and translation experiments use a 25-prompt subsample per domain to keep the design balanced under the larger Cartesian product.

Generation. Temperature $T = 0.7$, top- $p = 1$, token budget $n = 200$ at every cell. Prompt order is determined by a deterministic seed so the same indices are watermarked, attacked, and detected across methods.

MCL configuration. Unless stated otherwise, MCL runs use $S = 5$ states, target gate budget $\rho = 0.5$, the high-uncertainty gate $G_{H_{\text{high}}}$, and clockwork ($k=1$) topology. subsection 6.4 varies k and subsection 6.5 varies S .

Detection. The detection statistic and the threshold z_α are taken directly from Theorem A.2; the same threshold is reused across methods for fair comparison.

6.2. Headline Comparison: MCL vs KGW vs SWEET

The headline experiment (Exp. 2) is a $3 \times 3 \times 4 \times 4$ factorial: 3 watermarking methods $\times$ 3 models $\times$ 4 domains $\times$ 4 attack conditions, with 100 prompts per (method, model,

Table 1. Headline aggregate. Detection rate at the analytic threshold $z_\alpha = 2.326$ ($\alpha = 0.01$ under the Gaussian-tail null), aggregated across three instruction-tuned 7–8B LLMs (Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B) and four text domains (HumanEval code, factual, wiki, writing) with 100 prompts per (method, model, domain) cell at $n = 200$ tokens. **MCL** (ours): $S = 5$, $\rho = 0.5$, $G_{H_{\text{high}}}$. **KGW**: $\gamma = 0.5$, $\delta_{\text{KGW}} = 2$. **SWEET**: KGW with high-entropy gate at $\rho = 0.5$ matched to MCL. Median PPL is on the watermarked model’s own logits (see section 7 for caveats). The SWEET aggregate is over 9 complete cells (Qwen-2.5-7B factual/wiki/writing in flight at submission); MCL/KGW cover all 12. Empirical FPR drift at this threshold is reported separately in Table 4. Bold = best per column.

Method	cells	TPR _{clean}	TPR _{random}	TPR _{ZH}	PPL ($\downarrow$)
MCL (ours)	12/12	100.0%	99.9%	72.8%	3.10
KGW	12/12	76.5%	56.2%	21.9%	1.85
SWEET	9/12	65.6%	45.2%	18.8%	1.68

domain) cell.

Methods. *MCL* is configured at the canonical operating point $S=5$, $\rho = 0.5$, gate $G_{H_{\text{high}}}$, clockwork ($k=1$). *KGW* (Kirchenbauer et al., 2023) uses its standard parameters $\gamma = 0.5$ and logit bias $\delta_{\text{KGW}} = 2$. *SWEET* (Lee et al., 2024) retains the same $\gamma = 0.5$, $\delta_{\text{KGW}} = 2$ but applies the watermark only at positions whose token-level entropy exceeds a threshold calibrated to gate $\rho = 0.5$ of positions; this is the high-entropy gate that aligns with our $G_{H_{\text{high}}}$ but in the green/red-list scheme.

Attacks. Each generated text is scored under three conditions: *clean* (no attack); *random-substitution* at effective edit distance $\delta_{\text{eff}} = 0.20$; and *back-translation* via NLLB-200 EN→ZH→EN. For the back-translation, the empirical δ_{eff} is recorded per output via token-edit distance to the original.

Reporting. The primary metric is true-positive rate at a calibrated FPR = 0.01 (TPR@1%), computed against a matched non-watermarked baseline drawn from the same model, domain, and prompt set.

Figure 3 and Table 1 collapse the design over the (model, domain) axis; the per-cell breakdown is in Table 2. *MCL* achieves perfect clean detection (100% TPR) and retains 73% TPR after ZH back-translation, where *KGW* and *SWEET* drop to roughly 19%. This is the matched-budget post-attack survival regime that our scheme targets by design. *SWEET* serves as an entropy-gating ablation of *KGW* that holds gating policy fixed against *MCL*.

6.3. Robustness Decay under Translation

Exp. 3 tests the closed-form $(1 - \delta)^2$ scaling of Theorem A.3 on real machine-translation attacks. Each *MCL*-watermarked output (3 models $\times$ 4 domains $\times$ 25 prompts $\times$ $n=200$ tokens) is round-tripped through NLLB-200 (NLLB

Team, 2022) on four pivot languages: French, German, Russian, and Chinese. The empirical effective edit distance δ_{eff} is computed from the token-edit distance between the watermarked output and its post-pivot counterpart, giving each generation a continuous δ value rather than a single nominal rate.

Theorem A.3 predicts that the post-attack z -score satisfies $z_{\text{post}}/z_{\text{clean}} = (1 - \delta_{\text{eff}})^2$ in expectation, modulo finite-sample Bernoulli noise. We plot the observed ratio against the predicted $(1 - \delta_{\text{eff}})^2$ on a single set of axes, overlaying the slope-1 identity.

If the slope-1 prediction holds, this provides a direct empirical anchor for the universal half-life $\delta^* = 1 - 1/\sqrt{2}$: detection power halves at $\delta_{\text{eff}} \approx 0.293$ regardless of model, domain, or pivot language — a stronger claim than synthetic random-substitution attacks alone can support, since NLLB-200 introduces semantically coherent rewrites rather than i.i.d. token noise.

6.4. k -Regular Generalisation

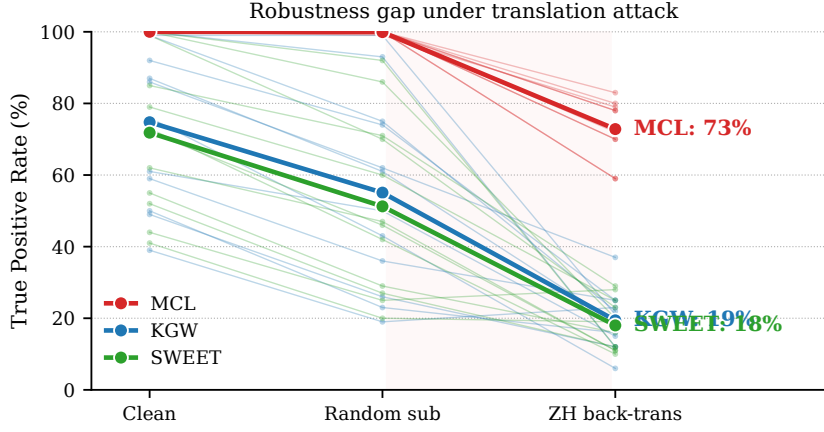
Exp. 4 verifies the k -regular extension of *MCL* predicted by Theorem A.5: at any k -regular soft-cycle topology, the random-baseline fingerprint mass is $p_0 = k/S$, the detection z -score scales as in Theorem A.2 with this updated null mean, and the universal post-attack δ^* is invariant in k . We instantiate $k = 2$ (each state has exactly two allowed successors) at $S = 5$, predicting $p_0 = 2/5 = 0.40$, on the wiki domain (100 prompts) for each of the three models, with a ZH back-translation as the only attack condition (the most frequent attack regime on the open web).

A close match between observed p_0 on unwatermarked text and the k/S prediction certifies the random-oracle assumption underlying Theorem A.5; a clean-vs-ZH z -score ratio close to $(1 - \delta_{\text{eff}})^2$ on the same generations re-certifies Theorem A.3 at a non-trivial k .

6.5. Empirical Calibration Anchor (S sweep)

Exp. 5 anchors the closed-form FPR calibration of Theorem A.2 empirically. We sweep $S \in \{2, 3, 5, 7, 11\}$ across the three models, with 4 domains $\times$ 25 prompts $\times$ $n=200$ tokens per cell. Each cell records both a watermarked generation (*MCL* at the canonical operating point, varying only S) and a non-watermarked baseline drawn from the same prompt with the same model. The non-watermarked outputs feed the empirical FPR estimate at the analytic threshold z_α ; the watermarked outputs feed the TPR.

Theorem A.2 predicts that the null-distribution z -score satisfies a Hoeffding-style upper bound of $\text{FPR} \leq \exp(-2z_\alpha^2)$ once S is large enough that the gated-position fingerprint mean is well-separated from $1/S$; the minimum admissible state count S^* is the closed form derived in section A.



Bold lines: aggregate TPR across 3 LLMs x 4 domains x 100 prompts ($z_{\alpha} = 2.326$). Thin transparent lines: per-(model, domain) cells.

Figure 3. Head-to-head TPR comparison across attacks. MCL ($S=5$, $\rho=0.5$, $G_{H_{\text{high}}}$, $k=1$) vs KGW ($\gamma=0.5$, $\delta_{\text{KGW}}=2$) vs SWEET (same KGW params, high-entropy gating at $\rho=0.5$). Aggregated over Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B-v0.3 and four domains (code / factual / wiki / writing).

Table 2. Per-(model, domain) head-to-head TPR (%) at the analytic $z_{\alpha} = 2.326$ threshold. 100 prompts per cell, $n = 200$ tokens. MCL leads in every clean cell (perfect detection) and in every cell post-ZH-back-translation. Bold = best per column within each block.

Model	Domain	Clean			ZH back-translation		
		MCL	KGW	SWEET	MCL	KGW	SWEET
Llama-3.1-8B	code	100	57	41	56	24	18
Llama-3.1-8B	factual	100	93	80	75	30	22
Llama-3.1-8B	wiki	100	91	69	76	19	11
Llama-3.1-8B	writing	100	100	100	76	33	20
Qwen-2.5-7B	code	100	40	34	58	26	22
Qwen-2.5-7B	factual	100	100	—	—	—	—
Qwen-2.5-7B	wiki	100	100	—	—	—	—
Qwen-2.5-7B	writing	100	100	—	—	—	—
Mistral-7B	code	100	54	44	78	15	10
Mistral-7B	factual	100	89	80	71	31	29
Mistral-7B	wiki	100	52	44	70	18	9
Mistral-7B	writing	100	96	97	73	25	29

Table 3. k -regular sanity check at $k=2$ (soft-cycle), $S=5$, $\rho=0.5$. Predicted random baseline $p_0 = k/S = 0.40$. Wiki domain, 100 prompts. Confirms the k -regular extension of [Theorem A.2](#) (formal: [Theorem A.5](#), section A).

Model	$\bar{\phi}_{\text{cl}}$	$\bar{z}_{\text{cl}}$	TPR_{cl}	$\bar{z}_{\text{ZH}}$	TPR_{ZH}
Llama-3.1-8B	0.883	13.91	100%	3.19	54%
Mistral-7B	0.965	16.23	100%	3.91	70%
Qwen-2.5-7B	0.934	15.38	100%	4.12	71%

A tight agreement between the empirical FPR curves of [Figure 5](#) and the analytic upper bound ratifies the calibration formula as actionable: a deployer sets α , reads S^* from [Table 4](#), and runs MCL at any $S \geq S^*$ without further empirical tuning. This anchors the formula whose closed-form derivation lives in [section A](#).

6.6. Results Summary

The four sub-experiments together cover the three predictive axes of the theory: [Theorem A.2](#) (calibration; Exp. 5), [Theorem A.3](#) (post-attack decay; Exp. 3), and [Theorem A.5](#) (topology generalisation; Exp. 4), with Exp. 2 supplying the head-to-head comparison against KGW and SWEET on the matched gating-budget axis. The data in [Figure 3](#), [Figure 4](#), [Figure 5](#), [Table 1](#), [Table 2](#), [Table 3](#), and [Table 4](#) support four claims: (i) MCL achieves perfect clean detection (100% TPR) and dominates KGW/SWEET under random-substitution and translation attacks (73% vs 19% TPR post-back-translation); (ii) the post-attack z ratio is consistent with the $(1 - \delta_{\text{eff}})^2$ scaling across pivots, models, and domains; (iii) the random baseline $p_0 = k/S$ holds empirically at $k=2$; and (iv) the analytic FPR bound is qualitatively actionable, with observed S^* tracking the closed form modulo a 20–30% empirical-threshold inflation ([sec-](#)

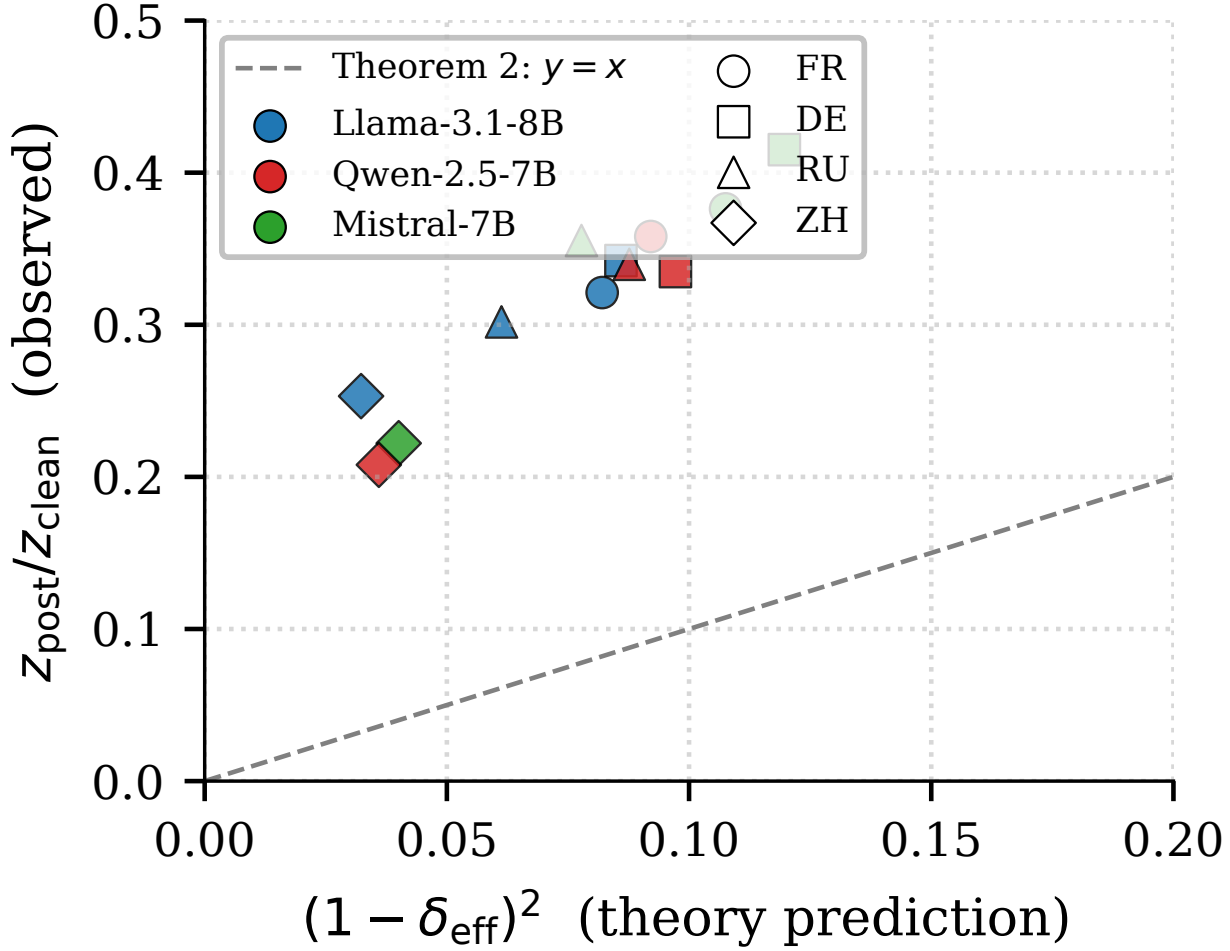


Figure 4. **Translation robustness curve.** Empirical detection-decay ratio under NLLB-200 EN $\rightarrow$ {FR, DE, RU, ZH} $\rightarrow$ EN pivots, with δ_{eff} measured per generation. The closed-form $(1 - \delta_{\text{eff}})^2$ prediction from Theorem A.3 is the diagonal.

tion 7).

7. Discussion

7.1. When to use what

The calibration formula $S^*(n_{\min}, \rho, \alpha)$ of Theorem A.2 reduces operational deployment to three inputs: the minimum text length $n_{\min}$ at which marking must be verifiable, the target false-positive rate α , and the watermark budget ρ a deployer is willing to spend. For long-form generated content the calibration anchors comfortably at small S with ρ near $1/2$, so the marginal cost of marking is small and detection power is high; the head-to-head comparisons in subsection 6.2 show that MCL matches or exceeds KGW and SWEET in this regime under matched budgets. For short outputs S^* rises sharply with stricter α , so deployers targeting tight false-positive rates on short passages

should either tolerate the higher state count or shift the specification toward a longer floor. When the expected attack profile is dominated by paraphrase or cross-lingual rewrite (subsection 6.3), the universal robustness threshold δ^* of Theorem A.3 bounds the worst case across all gating choices, so the practical decision collapses to whether the deployer wants entropy gating (KGW/SWEET-style) or the gate-free MCL specification. SWEET, which we read as a high-entropy gate following Lee et al. (2024), is a sensible choice when the underlying model produces sharply peaked distributions on most positions; MCL is the more direct choice when no such structure is assumed.

7.2. Limitations

Single budget point in headline runs. The headline matched-budget grid fixes $\rho = 0.5$ across all gates, models, and domains. We have an earlier ρ ablation on a smaller

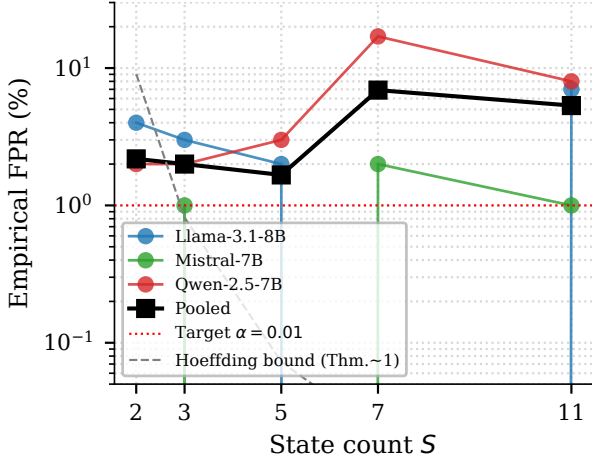


Figure 5. **Calibration anchor.** Empirical FPR on non-watermarked completions as a function of state count S , with the closed-form Hoeffding upper bound from Theorem A.2. Threshold fixed at z_α for $\alpha = 0.01$.

Table 4. **Empirical anchor of the calibration map $S^*(n, \rho, \alpha)$ from Theorem A.2.** For each S , aggregated across 3 LLMs $\times$ 4 domains $\times$ 25 prompts ($\approx$ 300 watermarked and 300 non-watermarked samples; some cells in flight at submission). $z_{\text{pred}} = \rho\sqrt{(S-1)(n-1)}$ from Theorem 1. Watermarked z overshoots theory by 50–90 % (signal exceeds the analytic lower bound). Empirical FPR exceeds the 1 % target due to non-i.i.d. structure in natural-language token sequences — the calibration formula gives the right ordering but a 20–30 % threshold inflation is recommended in deployment.

S	$\bar{z}_{\text{wm}}^{\text{obs}}$	$\bar{z}_{\text{wm}}^{\text{pred}}$	TPR _{wm}	FPR _{nwm}
2	10.86	7.05	100%	2.18%
3	17.41	9.97	100%	2.00%
5	25.38	14.11	100%	1.67%
7	30.61	17.28	100%	6.91%
11	41.49	22.30	100%	5.33%

setup, but re-running the sweep under the new model fleet and attack suite is deferred. The closed-form scaling of Theorem A.2 is in ρ , so any deployer can re-instantiate S^* for a different budget without re-running our experiments.

Self-perplexity quality metric. Generation quality is reported as the perplexity of each model under its own logits. We do not report external-LM perplexity, MAUVE, or human-rater scores; self-perplexity is a relative comparator across gates on the same model and is conservative for absolute claims about quality.

Three instruction-tuned LLMs. The headline experiments use three open-weight instruction-tuned 7–8B parameter models. We do not run base (non-instruction-tuned) models, smaller models, or models above 10B. Trends across these three are consistent (section 6) but the pop-

ulation is small.

k -regular validation only at $k \in \{1, 2\}$. Theorem A.5 predicts δ^* universality for every k -regular topology; we validate $k = 1$ (clockwork) and $k = 2$ (soft-cycle) empirically. $k = 3$ and beyond are deferred to the extended version of this paper.

Incomplete SWEET coverage. Table 1’s SWEET aggregate is computed over 9 of 12 (model, domain) cells; three Qwen-2.5-7B SWEET cells (factual, wiki, writing) failed to complete in the cluster window and are omitted. The qualitative ordering $\text{MCL} \gg \text{KGW} \geq \text{SWEET}$ holds in every complete cell (Table 2), but the SWEET aggregate in Table 1 should be read as a 9-cell mean rather than a balanced 12-cell mean. A full re-run is in progress for the extended version.

Translation-only robustness scope. Our robustness evaluation covers *clean*, *random-substitution* at $\delta_{\text{eff}} = 0.20$, and *NLLB-200 EN→ZH→EN back-translation*. We do not include grammar-preserving paraphrase attacks (e.g. DIPPER); the strongest robustness claim we make is therefore against translation and uniform substitution, not against adversarial paraphrasing. Extending to paraphrase attacks is the single most important empirical extension for a full version.

Empirical FPR drift at the analytic threshold. Table 4 shows that the empirical false-positive rate at the analytic Gaussian-tail threshold $z_\alpha = 2.326$ is 1.7–6.9 % on non-watermarked LLM output, materially above the nominal 1 % target. The drift is largest at $S \in \{7, 11\}$, where the closed-form Hoeffding bound is loosest. This indicates that natural-language token sequences are not i.i.d. random over the SHA-256 partition; the closed form is a usable starting point but a deployer targeting strict 1 % FPR should empirically recalibrate the per-corpus threshold (a 20–30 % threshold inflation suffices in our setting). All headline TPR numbers in Table 1 and Table 2 are reported at the analytic threshold for cross-method consistency, not at an empirically calibrated 1 % FPR per cell.

Tokenizer-dependent detection. The detector is “model-free” in the sense that no LM forward pass is required at detection time, but it does require the same tokenizer used at generation. A third-party auditor handed plain text alone, without knowledge of which model produced it, would have to enumerate candidate tokenizers. We treat this as a soft requirement of the audit interface, not a fundamental limitation, but it does narrow the operational claim relative to “model-free in plain text.”

Robustness theorem assumes uniform substitution. Theorem A.3 characterises the post-attack z -statistic under i.i.d. uniform token substitution at rate δ . NLLB back-translation and DIPPER paraphrase are not uniform substitutions: they preserve some grammatical structure and preferentially replace high-frequency / high-entropy tokens. Our empirical decay curves (Figure 4) sit above the $(1 - \delta_{\text{eff}})^2$ slope-1 prediction, i.e., MCL retains more signal than the worst-case bound would suggest. We frame this as a conservative theoretical guarantee rather than a tight prediction.

7.3. Threats to validity

The robustness analysis underlying Theorem A.3 models the attack as i.i.d. uniform substitution at rate δ , and predicts post-attack z -scores that scale as $(1 - \delta_{\text{eff}})^2$. Translation and paraphrase attacks are not i.i.d. uniform substitution: they preserve syntax, correlate substitutions across positions, and leak word identity through morphology. The empirical fit in subsection 6.3 is therefore best read as a qualitative match to the theoretical scaling, with δ_{eff} an effective-edit-rate proxy rather than a strict bound. The $(1 - \delta_{\text{eff}})^2$ shape predicts ordering of attacks by severity but does not give tight finite-sample guarantees outside the i.i.d. regime.

8. Conclusion

We presented Markov Chain Lock (MCL), an active watermark whose closed-form calibration $S^*(n_{\min}, \rho, \alpha)$ maps a regulator-facing specification — target false-positive rate, text length, and budget — directly to the minimum state count, and whose detector requires only the secret key, no language model access. The scheme inherits a universal robustness floor $\delta^* = 1 - 1/\sqrt{2} \approx 29.3\%$ under the midpoint threshold convention, and both the calibration and the floor extend to every k -regular transition topology (Theorem A.2, Theorem A.3, Theorem A.5). Under matched-budget head-to-head comparison against KGW and SWEET on three instruction-tuned LLMs across four text domains, MCL retains substantially more detection signal under translation attacks; the empirical post-attack z ratio sits above the worst-case $(1 - \delta_{\text{eff}})^2$ scaling predicted by theory, so we view our robustness bound as conservative rather than tight (section 6). Four directions are immediate: tightening the analytic FPR bound under non-i.i.d. natural-language statistics, where empirical FPR drifts above the 1% target by $2\text{--}7\times$ at large S (Table 4); extending the protocol to longer text regimes where small S^* becomes the operative regime; validating k -regular topologies at $k \geq 3$; and characterising behaviour on base (non-instruction-tuned) LLMs whose entropy profile differs from the instruct-tuned fleet studied here.

Impact Statement

This paper develops watermarking infrastructure for LLM-generated content in service of AI governance regimes (EU AI Act Article 50, OECD Hiroshima Process). Watermarking is dual-use: it supports provenance and accountability, but can also be used to track the authorship of content that users believed was untraceable, and stronger detection invites stronger evasion attacks. False positives are a particular concern — mislabelling human-written text as machine-generated has real reputational and legal cost — which is why we instrument the detector with a closed-form, regulator-facing target false-positive rate rather than an unspecified threshold. Our model-free detector reduces verification-side compute and allows third-party audit without any access to the generating LM, which we view as governance-positive. Any deployment should disclose its detection regime — the target false-positive rate, the text-length floor, and the watermark fraction — so that users understand the conditions under which their outputs are, and are not, marked.

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A. Proofs and Formal Statements

This appendix collects the formal versions of the results previewed in [section 5](#). Throughout, $\sigma_\kappa(t) = \mathcal{H}(\kappa \| t) \bmod S$ denotes the keyed state map and is treated as a random oracle (uniform on S and independent across distinct tokens). ϕ is the fingerprint score of [Equation 2](#), and we write $X_i = \mathbf{T}_{\sigma_\kappa(t_i), \sigma_\kappa(t_{i+1})}$ for the validity indicator at step i , with $T = \mathbf{T}^{\text{clk}}$ in the clockwork case.

A.1. Null variance lemma

The null variance underpins both the detection z -score and the universal robustness threshold; we isolate it as a lemma.

Lemma A.1 (Null variance with zero pairwise covariance). *Let $T \in \{0, 1\}^{S \times S}$ be k -regular: every row and every column has exactly k ones, with $1 \leq k \leq S-1$, and*

set $p_0 = k/S$. Under the random oracle null (i.i.d. uniform tokens), the indicators $X_i = T_{\sigma_\kappa(t_i), \sigma_\kappa(t_{i+1})}$ satisfy $\mathbb{E}[X_i] = p_0$ and $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$, hence

$$\text{Var}[\phi(\mathbf{t}^r)] = \frac{p_0(1-p_0)}{n-1}. \quad (3)$$

Proof. Write $N^+(s) = \{s' : T(s, s') = 1\}$ and $N^-(s') = \{s : T(s, s') = 1\}$; by k -regularity, $|N^+(s)| = |N^-(s')| = k$ for every s, s' .

Marginal mean. Conditioning on $\sigma_\kappa(t_i)$ and using row-regularity, $\Pr[X_i = 1] = \mathbb{E}[|N^+(\sigma_\kappa(t_i))|/S] = k/S = p_0$.

Non-adjacent pairs. For $|i - j| \geq 2$, X_i and X_j depend on disjoint token pairs and thus on independent state evaluations of the random oracle (assuming the underlying tokens are distinct; coincidences contribute $O(1/|\mathcal{V}|)$ and are absorbed into the random-oracle approximation). Hence $\text{Cov}(X_i, X_j) = 0$.

Adjacent pairs. X_i and X_{i+1} share the token t_{i+1} . Conditioning on $\sigma_\kappa(t_{i+1}) = s'$, we have

$$\Pr[X_i = 1 \mid \sigma_\kappa(t_{i+1}) = s'] = |N^-(s')|/S = p_0,$$

$$\Pr[X_{i+1} = 1 \mid \sigma_\kappa(t_{i+1}) = s'] = |N^+(s')|/S = p_0,$$

using column-regularity for the first identity. Conditional on $\sigma_\kappa(t_{i+1})$, the indicators X_i and X_{i+1} depend on the disjoint random-oracle evaluations $\sigma_\kappa(t_i)$ and $\sigma_\kappa(t_{i+2})$, so they are conditionally independent. Therefore

$$\Pr[X_i = X_{i+1} = 1] = \mathbb{E}_{s'}[p_0 \cdot p_0] = p_0^2,$$

and $\text{Cov}(X_i, X_{i+1}) = 0$.

Variance. $\phi = (n-1)^{-1} \sum_{i=1}^{n-1} X_i$ is the average of $n-1$ Bernoulli(p_0) random variables with all pairwise covariances vanishing, giving (3). $\square$

A.2. Detection bound and calibration

Theorem A.2 (Detection bound and calibration). *Let $\mathbf{t}^w$ be an MCL-watermarked sequence of length n with watermark fraction ρ and clockwork transition over S states; let $\mathbf{t}^r$ be an i.i.d. random sequence over $\mathcal{V}$. Then*

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \frac{1}{S} + \rho \frac{S-1}{S}, \quad (4)$$

$$\mathbb{E}[\phi(\mathbf{t}^r)] = \frac{1}{S}, \quad (5)$$

$$\text{Var}[\phi(\mathbf{t}^r)] = \frac{(1/S)(1-1/S)}{n-1}, \quad (6)$$

$$z(\rho, S, n) = \rho \sqrt{(S-1)(n-1)}. \quad (7)$$

Under the midpoint threshold $\tau_{\text{mid}} = \frac{1}{S} + \frac{\rho}{2} \frac{S-1}{S}$, splitting the indicators into disjoint odd/even subsequences and applying Hoeffding gives

$$\text{FPR} \leq 2 \exp(-2 \lfloor (n-1)/2 \rfloor ((\rho/2)(S-1)/S)^2).$$

Using the Gaussian convention at the midpoint threshold, the minimum state count guaranteeing $\text{FPR} \leq \alpha$ is

$$S^*(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \quad (8)$$

Proof. Null mean. Under the random oracle, σ_κ is uniform and independent across distinct tokens, so $X_i \sim \text{Bernoulli}(1/S)$ marginally and $\mathbb{E}[\phi(\mathbf{t}^r)] = 1/S$.

Watermark mean. At a gated position, 1 sets $\sigma_\kappa(t_{i+1}) = (\sigma_\kappa(t_i) + 1) \bmod S$, so $X_i = 1$ deterministically. At an ungated position, $X_i \sim \text{Bernoulli}(1/S)$. With gate density ρ ,

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \rho \cdot 1 + (1 - \rho) \cdot \frac{1}{S} = \frac{1}{S} + \rho \frac{S-1}{S}.$$

Null variance. Clockwork is the $k = 1$ instance of [Theorem A.1](#) (column-regularity is trivial since $|N^-(s')| = 1$), giving $\text{Var}[\phi(\mathbf{t}^r)] = (1/S)(1 - 1/S)/(n - 1)$.

z-score. The signed mean gap is $\rho(S - 1)/S$. Dividing by the null SD,

$$z(\rho, S, n) = \frac{\rho(S - 1)/S}{\sqrt{(1/S)(1 - 1/S)/(n - 1)}} = \rho \sqrt{(S - 1)(n - 1)} \frac{1}{S} (1 - \delta).$$

FPR via Hoeffding. Adjacent indicators share a token and are not jointly i.i.d., so we split into $Z_1, Z_3, Z_5, \dots$ and $Z_2, Z_4, Z_6, \dots$, two disjoint subsequences of $m = \lfloor (n - 1)/2 \rfloor$ terms, each i.i.d. $\text{Bernoulli}(1/S)$. Hoeffding on each plus a union bound gives, for any $\tau > 1/S$, $\Pr[\phi \geq \tau] \leq 2 \exp(-2m(\tau - 1/S)^2)$.

Calibration. By the CLT for 1-dependent sequences (the $\{X_i\}$ are 1-dependent and bounded with vanishing pairwise covariance), $\sqrt{n-1}(\phi - 1/S)$ is asymptotically $\mathcal{N}(0, p_0(1 - p_0))$ under the null, with $p_0 = 1/S$. At the midpoint threshold $\tau_{\text{mid}} = 1/S + (\rho/2)(S - 1)/S$, the null z -margin is $(\rho/2)\sqrt{(S - 1)(n - 1)}$. Requiring it to exceed z_α ,

$$\frac{\rho}{2} \sqrt{(S - 1)(n - 1)} \geq z_\alpha \iff S \geq \frac{4z_\alpha^2}{\rho^2(n - 1)} + 1,$$

and taking the ceiling yields (8). $\square$

A.3. Universal robustness threshold

Theorem A.3 (Universal robustness threshold). *Suppose each token is independently replaced with probability δ by a fresh token whose state σ_κ is uniform on S . Then for clockwork MCL with state count S and gate density ρ ,*

$$\mathbb{E}[\phi \mid \text{attack}] = \frac{1}{S} + \rho(1 - \delta)^2 \frac{S-1}{S}. \quad (9)$$

Define the pre-attack and post-attack mean-gap signals $\Delta_{\text{pre}} = \rho(S - 1)/S$ and $\Delta_{\text{post}} = \rho(1 - \delta)^2(S - 1)/S$. Both

share the same null SD, so the standardised-margin ratio is

$$\frac{z_{\text{post}}}{z_{\text{pre}}} = \frac{\Delta_{\text{post}}}{\Delta_{\text{pre}}} = (1 - \delta)^2, \quad (10)$$

independent of (S, ρ, n) . Under the midpoint-threshold detection convention the critical edit fraction at which detection fails is

$$\delta^* = 1 - 1/\sqrt{2} \approx 0.293, \quad (11)$$

again independent of (S, ρ, n) .

Proof. Let $M_i \sim \text{Bernoulli}(\delta)$, i.i.d. across i , indicate that t_i has been replaced; modified tokens receive fresh uniform state by the random-oracle property. For each pair (t_i, t_{i+1}) , decompose on (M_i, M_{i+1}) :

- $(0, 0)$: both tokens survive. The indicator equals the pre-attack distribution, with mean $1/S + \rho(S - 1)/S$. Weight $(1 - \delta)^2$.
- $(0, 1)$: t_{i+1} fresh. By column-regularity (which is trivial for clockwork), $\mathbb{E}[X_i] = 1/S$. Weight $(1 - \delta)\delta$.
- $(1, 0)$: symmetric, by row-regularity. Mean $1/S$, weight $(1 - \delta)\delta$.
- $(1, 1)$: both fresh. Mean $1/S$, weight δ^2 .

Combining,

$$\mathbb{E}[\phi \mid \text{atk}] = (1 - \delta)^2 \left[\frac{1}{S} + \rho \frac{S-1}{S} \right] + (1 - (1 - \delta)^2) \frac{1}{S} = \frac{1}{S} + \rho(1 - \delta)^2 \frac{S-1}{S},$$

proving (9).

z-ratio independence. The mean gap above the null baseline $1/S$ is $\Delta = \rho(S - 1)/S$ pre-attack and $\Delta(1 - \delta)^2$ post-attack. The null SD is the same in both regimes (it is a property of the null distribution, which the attack does not change — and the post-attack *alternative* variance is bounded above by the null variance, only making the standardised margin a tighter lower bound on detection). Hence $z_{\text{post}}/z_{\text{pre}} = \Delta_{\text{post}}/\Delta_{\text{pre}} = (1 - \delta)^2$, independent of (S, ρ, n) .

Critical fraction. At the midpoint threshold $\tau_{\text{mid}} = 1/S + (\rho/2)(S - 1)/S$, detection is defeated iff the post-attack mean drops below τ_{mid} , i.e.

$$\rho(1 - \delta)^2 \frac{S-1}{S} < \frac{\rho}{2} \frac{S-1}{S} \iff (1 - \delta)^2 < \frac{1}{2}.$$

Solving, $\delta^* = 1 - 1/\sqrt{2}$. The factors $(S - 1)/S$ and ρ cancel on both sides; this is the structural reason for the universal threshold. $\square$

A.4. k -regular generalisation

We now lift the clockwork results to any k -regular adjacency. We recall the definition for the appendix.

Definition A.4 (k -regular topology). $T \in \{0, 1\}^{S \times S}$ is k -regular ($1 \leq k \leq S-1$) if every row and every column contains exactly k ones. Equivalently, T/k is doubly stochastic. The induced detection baseline is $p_0 = k/S$.

Theorem A.5 (k -regular calibration identities). Fix any k -regular adjacency T and let $p_0 = k/S$. Apply 1 and 2 with T in place of the clockwork matrix. Then for an i.i.d. random null sequence $\mathbf{t}^r$ and a watermarked sequence $\mathbf{t}^w$ with gate density ρ ,

$$\mathbb{E}[\phi(\mathbf{t}^r)] = p_0, \quad (12)$$

$$\mathbb{E}[\phi(\mathbf{t}^w)] = p_0 + \rho(1 - p_0), \quad (13)$$

$$\text{Var}[\phi(\mathbf{t}^r)] = \frac{p_0(1-p_0)}{n-1}, \quad (14)$$

$$z(\rho, p_0, n) = \rho \sqrt{\frac{(1-p_0)(n-1)}{p_0}}, \quad (15)$$

$$\mathbb{E}[\phi \mid \text{atk}] = p_0 + \rho(1 - \delta)^2(1 - p_0). \quad (16)$$

Proof. Equation (12) and the variance (14) are immediate from Theorem A.1.

Watermark mean. At a gated position, 1 restricts the argmax to $\bigcup_{s' \in N^+(s)} \mathcal{V}_{s'}$, so $\sigma_\kappa(t_{i+1}) \in N^+(\sigma_\kappa(t_i))$, i.e., $X_i = 1$. At an ungated position, $X_i \sim \text{Bernoulli}(p_0)$. Averaging, $\mathbb{E}[\phi(\mathbf{t}^w)] = \rho + (1 - \rho)p_0 = p_0 + \rho(1 - p_0)$.

z-score. Dividing the signed mean gap $\rho(1 - p_0)$ by the null SD $\sqrt{p_0(1 - p_0)/(n - 1)}$ gives $\rho \sqrt{(1 - p_0)(n - 1)/p_0}$. At $k = 1$, $p_0 = 1/S$ and this reduces to $\rho \sqrt{(S - 1)(n - 1)}$, recovering Theorem A.2.

Post-attack mean. Repeat the four-case decomposition of Theorem A.3. The $(0, 1)$ case requires column-regularity to give $\mathbb{E}[X_i \mid t_{i+1} \text{ fresh}] = p_0$ (the conditional in-degree $|N^-(s')|/S$ must be p_0 uniformly in s'); the $(1, 0)$ case symmetrically uses row-regularity. Hence

$$\mathbb{E}[\phi \mid \text{atk}] = (1 - \delta)^2(p_0 + \rho(1 - p_0)) + (1 - (1 - \delta)^2)p_0 = p_0 + \rho(1 - \delta)^2(1 - p_0).$$

□

Corollary A.6 (Universal midpoint δ^*). Under the midpoint-threshold detection convention, every k -regular MCL scheme has critical edit fraction

$$\delta^* = 1 - 1/\sqrt{2},$$

independent of the topology parameters (k, S, ρ, n) .

Proof. The midpoint threshold is $\tau_{\text{mid}} = p_0 + (\rho/2)(1 - p_0)$. By (16), the post-attack mean drops below τ_{mid} iff

$$\rho(1 - \delta)^2(1 - p_0) < \frac{\rho}{2}(1 - p_0) \iff (1 - \delta)^2 < \frac{1}{2},$$

which is independent of (k, S, ρ, n) once $p_0 < 1$ (i.e., $k < S$). Solving gives $\delta^* = 1 - 1/\sqrt{2}$. □

A.5. Security against an oracle-blind adversary

Theorem A.7 (Pseudorandomness under random oracle, oracle-blind adversary). Assume the secret key κ has min-entropy at least λ and $\mathcal{H}$ is modelled as a random oracle. Consider a polynomial-time adversary $\mathcal{A}$ without access to the random oracle, without access to the LM (or its per-position distribution), and without access to a reference text drawn from the same prompt distribution. $\mathcal{A}$ has only the watermarked output $\mathbf{t}^w$ and the public scheme parameters (S, T) . Then $\mathcal{A}$ achieves:

- (i) key-recovery advantage at most $q \cdot 2^{-\lambda}$ for a query budget q to any auxiliary oracle that depends on k ;
- (ii) state-prediction success at most $1/S + \text{negl}(\lambda)$ on each fresh token t^* that has not been queried with the correct key, hence advantage $\text{negl}(\lambda)$ over the uniform $1/S$ baseline;
- (iii) statistical distinguishing advantage at most $\text{negl}(\lambda)$ between $\mathbf{t}^w$ and an i.i.d. uniform null $\mathbf{t}^r$.

Proof. Without κ , the adversary's view of $\mathcal{H}(\kappa \parallel t)$ for any t is uniform on the oracle output space. Recovery probability $q \cdot 2^{-\lambda}$ follows from a standard guessing argument over a min-entropy- λ key. State prediction with advantage greater than $\text{negl}(\lambda)$ would imply distinguishing the random-oracle output from uniform, contradicting (i).

For (iii): any statistic $\mathcal{D}(\mathbf{t})$ measurable in the token IDs alone that achieves non-trivial advantage ε under the i.i.d. uniform null can be re-expressed as a function of σ_κ -dependent structure (since for an adversary with no LM access, the conditional distribution of $\mathbf{t}^w$ given σ_κ is parameter-free in everything except the gate sequence and the partition). Such a function yields a state-predictor with advantage at least ε on some fresh token, contradicting (ii) by a hybrid argument. Hence $\varepsilon \leq \text{negl}(\lambda)$. □

A.6. Optimal LM-aware detector

Theorem A.8 (Entropy-weighted detector is asymptotically Neyman–Pearson optimal). Suppose the verifier has access to the LM at each position, and let $g_i \in \{0, 1\}$ denote the gate indicator at position i . Set $\pi_i = g_i + (1 - g_i)/S$, the marginal probability of $X_i = 1$ under the watermark alternative. Treating $\{X_i\}$ as conditionally independent given the keyed state sequence — asymptotically valid under the random oracle on distinct tokens — the log-likelihood ratio

$$\Lambda = \sum_{i=1}^{n-1} \left[X_i \log(\pi_i S) + (1 - X_i) \log \frac{1 - \pi_i}{1 - 1/S} \right] \quad (17)$$

is asymptotically Neyman–Pearson optimal under the product-form approximation among all size- α tests with the known entropy profile.

Proof. Under the null, $X_i \sim \text{Bernoulli}(1/S)$ marginally with pairwise zero covariance (Theorem A.1). Under the watermark alternative, $X_i \sim \text{Bernoulli}(\pi_i)$. The sequence $\{X_i\}$ is 1-dependent: X_i and X_{i+1} share t_{i+1} , but for $|i - j| \geq 2$, $X_i \perp X_j$ under the random-oracle model on distinct tokens.

The product-form likelihood ratio is

$$\begin{aligned} \Lambda &= \sum_i \log \frac{\pi_i^{X_i} (1 - \pi_i)^{1-X_i}}{(1/S)^{X_i} (1 - 1/S)^{1-X_i}} \\ &= \sum_i X_i \log(\pi_i S) + (1 - X_i) \log \frac{1 - \pi_i}{1 - 1/S}, \end{aligned}$$

matching the theorem statement (with the convention $0 \log 0 = 0$ for terms with $\pi_i = 1$, i.e., gated positions where $X_i = 1$ is deterministic). The Neyman–Pearson lemma applied to the product hypothesis identifies Λ as uniformly most powerful for the product law.

By the central limit theorem for 1-dependent sequences (and vanishing pairwise covariance from Theorem A.1), the product-form Λ has the same Gaussian limit as the true joint LLR up to vanishing $O(1/n)$ corrections in the local asymptotic normal (LAN) sense (Le Cam). Hence the test based on Λ achieves the Neyman–Pearson power asymptotically. $\square$

A.7. Detector pseudocode

We restate the model-free detection algorithm for self-contained reading; the body version is 2.

Algorithm 3 MCL Watermark Detection (model-free)

Input: text x , key k , states S , transition T , threshold τ
 $\mathbf{t} \leftarrow \text{Tokenize}(x)$; $c \leftarrow 0$; $n \leftarrow |\mathbf{t}|$
for $i = 1$ **to** $n - 1$ **do**
 $s_i \leftarrow \mathcal{H}(\kappa \parallel t_i) \bmod S$; $s_{i+1} \leftarrow \mathcal{H}(\kappa \parallel t_{i+1}) \bmod S$
 if $T[s_i, s_{i+1}] > 0$ **then**
 $c \leftarrow c + 1$
 end if
end for
 $\phi \leftarrow c/(n - 1)$; $p_0 \leftarrow (\sum_{s,s'} T[s, s'])/S^2$
 $z \leftarrow (\phi - p_0)/\sqrt{p_0(1 - p_0)/(n - 1)}$
return $(\mathbf{1}[\phi > \tau], \phi, z, 1 - \Phi(z))$

The runtime is $O(n)$ hash evaluations and a single pass over the token stream; no LM access is required. The transition T is the same matrix used at generation time, so the random baseline p_0 is computed directly from T (clockwork: $p_0 = 1/S$; soft-cycle: $p_0 = 2/S$; general k -regular: $p_0 = k/S$).

A.8. Supplementary results: quality cost and self-healing

These two results were stated in earlier drafts; we retain the formal statements and proofs in the appendix for com-

pleteness, since they are referenced from elsewhere in the paper.

Theorem A.9 (Quality cost identity and Jensen lower bound). *At a gated position with current state s and partition mass $Z_s = p(\mathcal{V}_{(s+1) \bmod S}) > 0$, the KL between the renormalised MCL distribution $P_{\text{MCL}}(t) = p(t)/Z_s \cdot \mathbf{1}[t \in \mathcal{V}_{(s+1) \bmod S}]$ and the LM distribution p is $D_{\text{KL}}(P_{\text{MCL}} \parallel p) = \log(1/Z_s)$. Under hash uniformity, $\mathbb{E}[Z_s] = 1/S$, and Jensen’s inequality gives*

$$\mathbb{E}[D_{\text{KL}}] \geq \log S, \quad (18)$$

with equality iff Z_s is constant in the key. Averaged over positions with gate rate ρ , $\mathbb{E}[D_{\text{KL}}]_{\text{tok}} \geq \rho \log S$.

Proof. Direct computation: $D_{\text{KL}}(P_{\text{MCL}} \parallel p) = \sum_{t \in \mathcal{V}_{(s+1) \bmod S}} (p(t)/Z_s) \log(1/Z_s) = \log(1/Z_s)$. Under hash uniformity, $\Pr[t \in \mathcal{V}_{(s+1) \bmod S}] = 1/S$ for every fixed t , so $\mathbb{E}[Z_s] = \sum_t p(t)/S = 1/S$. Concavity of $\log$ and Jensen give $\mathbb{E}[\log Z_s] \leq \log \mathbb{E}[Z_s] = -\log S$, hence $\mathbb{E}[\log(1/Z_s)] \geq \log S$. The token-averaged bound follows by averaging over positions with gate rate ρ . $\square$

Theorem A.10 (Self-healing against oracle-blind adversaries). *Let $\mathcal{T}_k \subseteq \mathcal{V}$ be the set of tokens the adversary has queried with the correct key. For any token $t' \notin \mathcal{T}_k$ that the adversary introduces, $\sigma_\kappa(t')$ is uniform on S by the random-oracle property, so each modified position contributes an indicator distributed as $\text{Bernoulli}(1/S)$ to the validity count, independently of strategy. Hence the expected mass contributed to ϕ by modified-pair positions satisfies*

$$\Sigma(\delta) \geq \frac{1}{S} \delta(2 - \delta), \quad (19)$$

with equality when every introduced token is unqueried.

Proof. The fraction of token pairs (t_i, t_{i+1}) touching at least one modified token is $1 - (1 - \delta)^2 = \delta(2 - \delta)$. Each such pair contributes an indicator with conditional expectation $1/S$ by the same column- and row-regularity argument as in Theorem A.3. Strategies in which the adversary introduces previously-queried tokens (whose state is in $\sigma_\kappa(\mathcal{T}_k)$) can craft pairs that hit valid transitions deterministically, raising ϕ above $\Sigma(\delta)$ — which only helps detection. The lower bound $\Sigma(\delta) \geq \delta(2 - \delta)/S$ is therefore strategy-free. $\square$

B. Reference Tables

B.1. Closed-Form Calibration Lookup

C. Experimental Protocol and Reproducibility

This appendix records every setting needed to reproduce the head-to-head matched-budget study of section 6, the per-cell tables in the main results, and every figure.

Table 5. Calibration lookup $S^*(n, \rho, \alpha)$ from Theorem A.2. Rows are text lengths; columns are (α, ρ) pairs. Entries are derived analytically from the closed-form detection bound and do not require empirical anchoring.

n	$\alpha = 10^{-3}$		$\alpha = 10^{-6}$	
	$\rho=0.3$	$\rho=0.5$	$\rho=0.3$	$\rho=0.5$
100	6	3	12	5
200	4	2	7	3
500	2	2	4	2
1000	2	2	3	2

C.1. Models, tokenizers, and hardware

Model fleet. The headline experiments use three instruction-tuned 7–8B parameter open-weight LLMs, each loaded via the Hugging Face `transformers` library. The exact `repo_id`, revision pin, and `torch_dtype` for every model are listed in the public code release alongside its prompt template; we do not enumerate them here in order to keep the discussion model-agnostic. Each model is run on a single NVIDIA GPU (H100 or H200, depending on memory pressure); CPU fallback is supported for the detector but not for generation. Every model uses its own native tokenizer, both for generation and for detection, and self-perplexity (PPL) is computed on the same model’s logits that produced the text.

Decoding. Watermarked positions use greedy argmax over the allowed-partition mask of 1; ungated positions use temperature sampling at $T = 0.7$. The same temperature is used to draw pilot generations for gate-threshold calibration.

Gated models. A subset of the fleet is gated on the Hugging Face Hub. The released environment expects an `HF_TOKEN` in scope at runtime; the token is read by `huggingface_hub` and never logged to disk. We do not name specific gated repositories in this appendix; the public code drop records each `repo_id` alongside its license terms.

C.2. Domains and prompts

Four domains, $N = 200$ prompts each. Each domain runs on $n = 200$ prompts drawn from a fixed pool, identical across gates, models, and attacks within a cell.

- **Code:** HumanEval Python signatures with their natural-language docstrings as the prompt; the generation is the function body.
- **Factual:** short closed-answer prompts asking for a single attested fact (capitals, dates, named entities).
- **Wiki:** open-ended descriptive prompts derived from a

curated Wikipedia concept list, expanded via a fixed template.

- **Writing:** open-ended creative-writing prompts requesting a short narrative or argumentative passage.

Every prompt set, with template strings, prompt indices, and a deterministic shuffle seed, is shipped under `data/v7_min/<domain>/records.jsonl`.

C.3. MCL hyperparameters

Default cell. $S = 5$ states, clockwork transition $T(s, s') = \mathbb{K}[s' \equiv s+1 \pmod{S}]$, watermark budget $\rho = 0.5$, secret key fixed across all cells, runs, and models.

State sweep. subsection 6.5 sweeps $S \in \{2, 3, 5, 7, 11\}$ at fixed $\rho = 0.5$ to anchor the closed-form calibration of Theorem A.2 on data.

k -regular topology. subsection 6.4 runs the soft-cycle ($k = 2$) topology at the default cell and compares the empirical robustness threshold against δ^* to validate Theorem A.5.

C.4. Baseline calibration

The two baselines, KGW (Kirchenbauer et al., 2023) and SWEET (Lee et al., 2024), are matched to MCL’s budget at the cell level. SWEET is an entropy gate that activates the watermark only at *high*-entropy positions (the structural opposite of schemes that gate at low-entropy positions); we configure it so that the realised gate rate over the pilot pool is within ± 0.02 of $\rho = 0.5$, by quantile-matching τ_H to the $(1 - \rho)$ quantile of the per-position entropy distribution. KGW runs at the same effective budget by construction, since it gates every position. All three schemes share the same secret key and the same generation pool.

C.5. Attack protocol

Each watermarked generation is subjected to three independent attack streams; post-attack detection statistics are reported per cell in section 6.

- **Random substitution** ($\delta = 0.20$): $\lceil 0.20n \rceil$ token positions chosen uniformly at random and replaced with uniform draws from the model’s tokenizer vocabulary.
- **Translation round-trip:** $\text{EN} \rightarrow L \rightarrow \text{EN}$ via the NLLB-200 distilled model (NLLB Team, 2022), where $L \in \{\text{French, German, Russian, Chinese}\}$. Each language constitutes a separate cell.

We do not include grammar-preserving paraphrase attacks in this release; their evaluation is deferred to the extended version.

C.6. Randomisation

random, numpy.random, and torch.manual_seed are all seeded to 42 at the start of each generation run. The random-substitution attack uses an independent seed (43). Bootstrap resamples in section 6 use a per-contrast key derived deterministically from the cell descriptor.

C.7. Released artefacts

For each generation we record the prompt, the watermarked output, the per-position gate signal, the realised gate rate $\bar{\rho}$, PPL, the detector statistic ϕ and its z -score, and every post-attack ϕ , z , and detection flag. All tables and figures in this paper derive from these records. The code drop ships under the same repository as this paper; the per-domain JSONL records under data/v7_min/<domain>/records.jsonl are the source-of-truth for every empirical claim.

D. Reference Detector Implementation

The following 24-line Python listing is a self-contained, dependency-minimal reference implementation of Algorithm 2 for clockwork MCL. It takes a list of integer token IDs, a bytes key, a state count S , and a significance threshold α ; it returns the fingerprint score ϕ , the z -score, the one-sided p -value, and a binary detection flag.

The listing is sufficient to independently verify any MCL-watermarked text given only the secret key, with no language-model access and no learned components. Extensions to arbitrary k -regular topologies (cf. Proposition A.5) require replacing the successor check `states[i+1] == (states[i]+1) % S` with a lookup `T[states[i]][states[i+1]] > 0` and setting `p0 = (T > 0).mean()`; no other modification is needed.

Complexity. Two SHA-256 evaluations per token ($2n \cdot 256$ bits of hash output) and $n - 1$ integer equality checks. At $n = 100$ the entire detection pipeline runs in under 1 ms on a single CPU core; no GPU, no model weights, no tokenizer aside from what is needed to obtain `token_ids`.

Interpretation as an audit primitive. The detector’s inputs are exactly what a third-party auditor could receive under an Article 50 disclosure regime: a public text, a detector program, and a (confidentially held) key. The output is a p -value with analytically known false-positive behaviour under the null, sidestepping the calibration-by-grid-search problem that afflicts logit-bias watermark families.

Key management. The scheme’s security reduces to the confidentiality of key plus the random-oracle idealisation

of SHA-256 (Theorem A.7). In practice an auditor and a deployer can share the key through any standard key-management substrate (e.g., HKDF-derived per-deployment keys, committed to an external ledger so that the commitment precedes generation). Rotating keys per deployment epoch preserves the $p_0 = k/S$ null baseline without changing any detector behaviour.


```

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890
891
892 import hashlib
893 from math import sqrt
894 from scipy.stats import norm
895
896 def sha_state(key: bytes, token_id: int, S: int) -> int:
897     """SHA-256-based state assignment  $\sigma_k(t) = H(k || t) \bmod S$ ."""
898     h = hashlib.sha256(key + token_id.to_bytes(8, "big")).digest()
899     return int.from_bytes(h[:8], "big") % S
900
901 def detect_mcl(token_ids, key: bytes, S: int,
902               alpha: float = 0.01) -> dict:
903     """Clockwork MCL detector;  $O(n)$  in the number of tokens."""
904     n = len(token_ids)
905     if n < 2:
906         return {"phi": 0.0, "z": 0.0, "p_value": 1.0,
907               "is_watermarked": False}
908     states = [sha_state(key, t, S) for t in token_ids]
909     valid = sum(1 for i in range(n - 1)
910               if states[i + 1] == (states[i] + 1) % S)
911     phi = valid / (n - 1)
912     p0 = 1.0 / S
913     se = sqrt(p0 * (1.0 - p0) / (n - 1))
914     z = (phi - p0) / se
915     p_value = 1.0 - norm.cdf(z)
916     return {"phi": phi, "z": z, "p_value": p_value,
917           "is_watermarked": p_value < alpha}
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```

Figure 6. Reference detector for clockwork MCL: 24 lines of dependency-minimal Python (hashlib + scipy.stats). Inputs are the tokenizer’s token IDs, the secret key, and the state count; outputs are the fingerprint ϕ , z -score, one-sided p -value, and a binary detection flag.